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Bank Market Power and Monetary Policy Transmission: Evidence from a Structural Estimation

Contents

1 Big picture

- **Question.** How much of monetary policy transmission through banks is driven by *bank market power*, relative to the traditional channels based on reserve requirements and capital regulation?
- **Setting.** U.S. commercial banks from 1994 to 2017. Banks take short-term deposits, make longer-term loans, face reserve and capital constraints, and compete imperfectly in both deposit and loan markets.
- **Core empirical strategy.** The paper estimates a dynamic structural banking model in two stages. First, it estimates deposit and loan demand using BLP-style methods. Second, it plugs those demand estimates into a dynamic bank problem and uses simulated minimum distance (SMD) to estimate funding frictions and operating costs.
- **Main channels.** The model isolates four channels: the reserve channel, the capital channel, the deposit-market-power channel, and the loan-market-power channel.
- **Main result.** Deposit market power is quantitatively very important. Removing deposit market power lowers the sensitivity of lending to the federal funds rate by about 31%, a larger effect than removing reserve requirements and slightly larger than removing capital regulation.
- **Capital channel result.** Capital regulation is also important. Removing the capital requirement lowers monetary transmission by about 24%.
- **Reserve channel result.** Reserve requirements contribute little in the sample period. Removing them changes the lending sensitivity by only about 9%.
- **Loan market power result.** Loan market power works in the opposite direction: because banks cut loan markups when rates rise, loan-market power *dampens* transmission. Removing it makes lending more rate-sensitive.
- **Reversal-rate result.** When the federal funds rate is below about 0.9%, further rate cuts can become contractionary because lower deposit spreads compress profits, lower retained earnings, tighten the capital constraint, and reduce lending.
- **Cross-sectional result.** Small banks are more sensitive to monetary policy mainly because external finance is more costly for them and because they have higher fixed operating costs.
- **Time-series result.** Transmission weakens over time. The sensitivity of lending to the federal funds rate falls from about 1.53 in the early subsample to 1.22 in the late subsample, reflecting lower average rates, changing market power, and better operating efficiency.
- **Why this paper matters.** The paper turns a largely qualitative literature into a quantitative decomposition. It shows that banking-industry structure—not just regulation—is central for understanding monetary policy transmission.

2 Data and measurement (key objects)

2.1 Main data sets

- **Call Reports.** Quarterly bank-level balance-sheet and income-statement data for U.S. commercial banks. These provide deposits, loans, interest income and expense, loan maturity, salary expenses, and fixed-asset-related expenses.

- **FDIC Summary of Deposits.** Annual branch-level deposit data, merged to the Call Reports to measure bank characteristics such as branch networks since 1994.
- **CRSP and NY Fed link table.** Used to construct publicly traded bank returns and connect them to local concentration measures.
- **FRED and other macro sources.** Effective federal funds rate, Treasury yields, aggregate corporate bond issuance, household holdings of cash, Treasury bills, and money-market funds, plus NBER recession dates.
- **FOMC dates.** Used for the event-study validation exercises around monetary policy announcements.

2.2 Sample and market definitions

- **Sample period.** 1994–2017.
- **Unit of analysis.** Bank-quarter for most balance-sheet moments; annual bank-level "market" observations for the BLP-style demand system.
- **Deposit market for estimation.** The U.S. national market, with each year treated as a separate market.
- **Loan market for estimation.** Also the national market, because many loan categories are not available at subnational levels.

2.3 Key variables from Table I

- **Deposit market share.** Bank deposits divided by the sum of deposits, cash, Treasury bills, and money-market funds in the U.S. economy.
- **Loan market share.** Bank loans divided by the sum of U.S. household debt, corporate debt, and corporate equity.
- **Deposit and loan rates.** Interest expense divided by deposits; interest income divided by loans.
- **Deposit spread.** Federal funds rate minus the bank deposit rate.
- **Loan spread.** Loan rate minus the corresponding five-year Treasury yield.
- **Reserve ratio.** A weighted average of statutory reserve requirements across deposit types.
- **Nonreservable borrowing share.** Nonreservable borrowing divided by total deposits.
- **Deposit-to-asset ratio.** Deposits divided by total assets.
- **Leverage.** Total assets divided by book equity.
- **Market-to-book ratio.** Market equity divided by book equity.

2.4 Summary statistics from Table II

- **Concentration.** The asset-weighted mean deposit market share is 3.519% and the asset-weighted mean loan market share is 1.368%, both far above the equal-weighted means, indicating a highly skewed size distribution with a few dominant banks.
- **Rates.** The asset-weighted mean deposit rate is 1.706%, while the asset-weighted mean loan rate is 5.935%.
- **Branch network heterogeneity.** The asset-weighted mean number of branches is 1,778, but the equal-weighted mean is only about 70, again showing the skewness in bank size.
- **Funding structure.** The deposit-to-asset ratio is about 0.707 asset-weighted, while the borrowing-to-deposit ratio is about 0.699 asset-weighted and only 0.136 equal-weighted.
- **Loan maturity.** Average loan maturity is about 3.43 years, so maturity transformation is quantitatively important.
- **Capital.** Mean book leverage is around 11.5, consistent with banks operating with thin equity cushions.

2.5 Stylized facts from Figure 1

- **Deposit spreads rise with the federal funds rate.** This means banks pass through rate hikes incompletely to depositors and charge larger markups when cash becomes a less attractive outside option.
- **Loan spreads fall with the federal funds rate.** This means banks compress loan markups when policy tightens, consistent with trying to cushion the fall in loan demand.
- **Interpretation.** The basic raw-data pattern already points to market power in both markets, with nontrivial state dependence at low rates.

3 Models and empirical specifications (all equations + interpretation)

3.1 Households and deposit demand

Households choose among cash, short-term bonds, and deposits at each bank. Household i's utility from option j is

$$u_{i,j} = \alpha_i^d r_j^d + \beta^d x_j^d + \xi_j^d + \varepsilon_{i,j}^d. \tag{1}$$

- **Interpretation.** Deposit demand is a discrete-choice problem. The rate r_j^d matters, but so do nonprice bank characteristics x_j^d, such as branches and employees per branch.
- **Heterogeneity.** Depositor rate sensitivity α_i^d is heterogeneous and uniformly distributed with mean α^d and standard deviation σ_α^d. This heterogeneity is essential for generating state-dependent deposit market power.

The implied logit-style deposit share for bank j is

$$s_j^d(r_j^d \mid f) = \sum_{i=1}^{I} \mu_i^d \frac{\exp\Big(\alpha_i^d r_j^d + \beta^d x_j^d + \xi_j^d\Big)}{\exp\Big(\alpha_i^d f + \beta^d x_{J+1}^d + \xi_{J+1}^d\Big) + \exp(\beta^d x_c^d + \xi_c^d) + \sum_{m=1}^{J} \exp(\alpha_i^d r_m^d + \beta^d x_m^d + \xi_m^d)}. \tag{2}$$

Aggregate deposits of bank j are then

$$D_j(r_j^d \mid f) = s_j^d(r_j^d \mid f) W. \tag{3}$$

- **Interpretation.** Deposit quantity equals market share times aggregate savings wealth W.
- **Outside options.** The denominator contains Treasury bills and cash, so the federal funds rate changes bank demand by changing the attractiveness of bank deposits relative to those outside assets.

3.2 Firms and loan demand

Firms choose between bank loans, bond issuance, and not borrowing. The long-term bond rate is the expected weighted average of future short rates plus default cost:

$$\bar{f}_t = \eta f_t + \mathbb{E}_t \left[\sum_{n=1}^{\infty} \eta (1-\eta)^n f_{t+n} \right]. \tag{4}$$

Firm i's payoff from choosing financing option j is

$$\pi_{i,j} = \alpha_i^l r_j^l + \beta^l x_j^l + \xi_j^l + \varepsilon_{i,j}^l. \tag{5}$$

The corresponding loan share for bank j is

$$s_j^l(r_j^l \mid \bar{f}) = \sum_{i=1}^{I} \mu_i^l \frac{\exp\Big(\alpha_i^l r_j^l + \beta^l x_j^l + \xi_j^l\Big)}{\exp\Big(\alpha_i^l(\bar{f} + \delta) + \beta^l x_{J+1}^l + \xi_{J+1}^l\Big) + \exp(\beta^l x_n^l + \xi_n^l) + \sum_{m=1}^{J} \exp(\alpha_i^l r_m^l + \beta^l x_m^l + \xi_m^l)}. \tag{6}$$

Aggregate bank lending is

$$B_j(r_j^l \mid \bar{f}) = s_j^l(r_j^l \mid \bar{f}) K. \tag{7}$$

- **Interpretation.** Loan demand equals market share times the total loan market K.
- **Economic mechanism.** Higher policy rates raise long-term funding costs and make the outside option of not borrowing or issuing bonds more attractive, which compresses bank loan demand.

3.3 Bank balance sheet, funding costs, and profits

Banks choose deposit and loan rates while transforming short-term liabilities into long-term loans. Deposit rates must satisfy the zero lower bound,

$$r^d_{j,t} \geq 0. \tag{8}$$

The present value of interest income on newly originated loans is

$$I_{j,t} = \sum_{n=0}^{\infty} \frac{(1-\eta)^n B_{j,t} r^l_{j,t}}{(1+\gamma)^n}. \tag{9}$$

Outstanding loans evolve according to

$$L_{j,t+1} = (1-\eta)(L_{j,t} + B_{j,t}). \tag{10}$$

Nonreservable borrowing carries a convex financing cost:

$$\mathcal{N}(N_t) = \left(f_t + \frac{\phi^N}{2} \frac{N_t}{D_t} \right) N_t. \tag{11}$$

Banks also incur linear deposit- and loan-servicing costs,

$$\mathcal{C}^d(D_t) = \phi^d D_t, \qquad \mathcal{C}^l(B_t + L_t) = \phi^l (B_t + L_t). \tag{12}$$

The balance sheet identity is

$$L_t + B_t + R_t + G_t = D_t + N_t + E_t, \tag{13}$$

and book equity evolves as

$$E_{t+1} = E_t + \Pi_t(1-\tau) - C_{t+1}. \tag{14}$$

Profits are

$$\begin{aligned} \Pi_t = I_t - (L_t + B_t)(\eta\delta_t + \phi^l) + G_t f_t - (r^d_t + \phi^d) D_t \\ - \left(f_t + \frac{\phi^N}{2} \frac{N_t}{D_t} \right) N_t - \psi \bar{E}. \end{aligned} \tag{15}$$

- **Interpretation of ϕ^N.** This is the key balance-sheet friction. When deposits fall, banks cannot replace them one-for-one with frictionless wholesale funding.
- **Interpretation of ϕ^d, ϕ^l, and ψ.** These govern marginal operating costs in deposits and loans and fixed net operating costs relative to steady-state equity.
- **Why this matters.** Profits feed directly into retained earnings, so market power and funding frictions interact with capital regulation.

3.4 Regulation and monetary-policy process

The bank cannot issue equity in the baseline model,

$$C_{t+1} \geq 0, \tag{16}$$

so inside equity can rise only via retained earnings. Banks also face a capital requirement and a reserve requirement,

$$E_{t+1} \geq \kappa(L_t + B_t), \qquad R_t \geq \theta D_t. \tag{17}$$

Monetary policy is an exogenous process for the real federal funds rate jointly evolving with loan charge-offs:

$$\begin{pmatrix} \ln \delta_{t+1} - \mathbb{E}(\ln \delta) \\ \ln f_{t+1} - \mathbb{E}(\ln f) \end{pmatrix} = \begin{pmatrix} \rho_\delta & \rho_{\delta f} \\ 0 & \rho_f \end{pmatrix} \begin{pmatrix} \ln \delta_t - \mathbb{E}(\ln \delta) \\ \ln f_t - \mathbb{E}(\ln f) \end{pmatrix} + \begin{pmatrix} \sigma_\delta & 0 \\ 0 & \sigma_f \end{pmatrix} \varepsilon_{t+1}. \tag{18}$$

- **Capital requirement.** This transmits monetary policy through the effect of rates on bank profits and capital accumulation.
- **Reserve requirement.** This makes deposits costly when rates rise because reserves earn zero in the baseline model.
- **Monetary-policy transmission.** Short rates affect funding costs directly and long rates indirectly through expectations.

3.5 Dynamic bank problem and equilibrium approximation

The bank solves

$$V(f_t, \delta_t, L_t, E_t \mid \Gamma_t) = \max_{\{r_t^l, r_t^d, G_t, N_t, R_t, C_{t+1}\}} \frac{1}{1+\gamma} \left[C_{t+1} + \mathbb{E}V(f_{t+1}, \delta_{t+1}, L_{t+1}, E_{t+1} \mid \Gamma_{t+1}) \right], \tag{19}$$

subject to the demand equations, balance-sheet constraints, regulatory constraints, and the law of motion for (f_t, δ_t).

- **Interpretation.** Banks internalize how today's pricing decisions affect deposits, loans, profits, future capital, and future ability to lend.
- **State variable problem.** The full industry state Γ_t is high-dimensional.

To make the model computationally feasible, the paper approximates the law of motion for other banks' choices by average loan and deposit rates conditional only on the current federal funds rate:

$$\exp\Big(\alpha_i^d \bar{r}_i^d(f_t) + \beta^d x^d + \xi^d\Big) \equiv \mathbb{E}\Big[\exp\Big(\alpha_i^d r_{j,t}^d + \beta^d x^d + \xi^d\Big) \mid f_t\Big], \tag{20}$$

$$\exp\Big(\alpha_i^l \bar{r}_i^l(f_t) + \beta^l x^l + \xi^l\Big) \equiv \mathbb{E}\Big[\exp\Big(\alpha_i^l r_{j,t}^l + \beta^l x^l + \xi^l\Big) \mid f_t\Big]. \tag{21}$$

- **Interpretation.** Banks forecast the rest of the industry using the current rate environment.
- **Accuracy check.** The paper reports R^2 values above 95% for deposit-rate forecasts and above 97% for loan-rate forecasts.

3.6 Static intuition: frictionless benchmark

In the static benchmark, banks solve

$$\Pi_j = \max_{\{r_j^l, r_j^d\}} \left[r_j^l B_j - r_j^d D_j - (B_j - D_j) f \right], \qquad r_j^d \geq 0. \tag{22}$$

- **Interpretation.** Without market power, funding frictions, or regulation, banks are pure pass-through entities.
- **Implication.** Deposit and loan rates both equal the federal funds rate.

3.7 Static intuition: imperfect competition and pass-through formulas

With imperfect competition, banks charge markups in both markets. The optimal loan and deposit rates become

$$r_j^l = f + \left[-\frac{\partial B_j / \partial r_j^l}{B_j} \right]^{-1} = f + \left[-\alpha^l \left(1 - \frac{S^l}{J} \right) \right]^{-1}, \tag{23}$$

$$r_j^d = f + \left[-\frac{\partial D_j / \partial r_j^d}{D_j} \right]^{-1} = f - \left[\alpha^d \left(1 - \frac{S^d}{J} \right) \right]^{-1}, \tag{24}$$

where $S^l = \sum_j s_j^l$ and $S^d = \sum_j s_j^d$.

- **Interpretation.** Loan rates equal marginal funding costs plus a markup; deposit rates equal the outside return minus a markdown.
- **Market structure.** Lower rate sensitivity and higher concentration imply bigger markups.

The response of markups to the policy rate is

$$\frac{\mathrm{d}(r_j^l - f)}{\mathrm{d}f} = -\frac{(S^l/J)(1 - S^l - s_{J+1}^l)}{(1 - S^l/J)^2 + S^l(1 - S^l)/J} < 0, \tag{25}$$

$$\frac{\mathrm{d}(f - r_j^d)}{\mathrm{d}f} = \frac{(S^d/J)(1 - S^d - s_{J+1}^d)}{(1 - S^d/J)^2 + S^d(1 - S^d)/J} > 0. \tag{26}$$

- **Interpretation.** When the policy rate rises, loan spreads fall but deposit spreads rise.
- **Economic intuition.** Higher rates make cash less attractive, so banks can squeeze depositors more; but they also weaken loan demand, so banks reduce loan markups to preserve borrowers.

The resulting quantity sensitivities in the simple static model are

$$\frac{\mathrm{d}\log B_j}{\mathrm{d}f} = \alpha^l \frac{(1-S^l/J)^2(1-S^l-s^l_{J+1})}{(1-S^l/J)^2+S^l(1-S^l)/J} < 0, \tag{27}$$

$$\frac{\mathrm{d}\log D_j}{\mathrm{d}f} = \alpha^d \frac{(1-S^d/J)^2(1-S^d-s^d_{J+1})}{(1-S^d/J)^2+S^d(1-S^d)/J} > 0. \tag{28}$$

- **Interpretation.** In the homogeneous-depositor static benchmark, loan quantities fall with rate hikes while deposit quantities rise.
- **Why the full model differs.** With heterogeneous depositors, higher rates can lower the average rate sensitivity of the bank's clientele and induce larger markups, so deposits can fall when rates rise.

3.8 Static intuition: balance-sheet frictions, reserve requirements, and capital regulation

With expensive wholesale funding, the static problem becomes

$$\Pi_j = \max_{\{r^l_j, r^d_j\}} \left[r^l_j B_j - r^d_j D_j - \mathcal{N}(N_j) \right], \qquad N_j = B_j - D_j. \tag{29}$$

With reserve requirements,

$$\Pi_j = \max_{\{r^l_j, r^d_j\}} \left[r^l_j B_j - r^d_j D_j - (B_j + R_j - D_j) f \right], \qquad R_j \geq \theta D_j. \tag{30}$$

With capital regulation,

$$\Pi_j = \max_{\{r^l_j, r^d_j\}} \left[r^l_j B_j - r^d_j D_j - (B_j - D_j) f \right], \qquad E_j + \Pi_j \geq \kappa B_j. \tag{31}$$

- **Balance-sheet friction.** Deposit shocks matter for lending because replacing deposits with nonreservable funding is costly.
- **Reserve channel.** Higher policy rates raise the opportunity cost of holding required reserves.
- **Capital channel.** Rate hikes can erode capital through maturity mismatch and reduce feasible lending.

3.9 Estimation equations used in the plug-in problem

After first-stage demand estimation, the fitted deposit-demand equation is

$$D_j(r^d_j \mid f) = \sum_{i=1}^{I} \mu^d_i \frac{\exp\!\left(\hat{\alpha}^d_i r^d_j + q^d_j\right)}{\exp(\hat{\alpha}^d_i f) + \exp(q^d_c) + \sum_{m=1}^{\hat{J}} \exp(\hat{\alpha}^d_i r^d_m + q^d_m)} W, \tag{32}$$

and the fitted loan-demand equation is

$$B_j(r_j^l \mid \bar{f}) = \frac{\exp\left(\hat{\alpha}^l r_j^l + q_j^l\right)}{\exp\left[\hat{\alpha}^l(\bar{f}+\delta)\right] + \exp(q_n^l) + \sum_{m=1}^{\hat{J}} \exp(\hat{\alpha}^l r_m^l + q_m^l)} K. \tag{33}$$

- **Interpretation.** The fitted quality terms q summarize the nonprice attractiveness of each product after the first-stage BLP estimation.
- **What is estimated in stage two?** Stage two takes these demand curves as given and estimates the balance-sheet-friction parameters that rationalize bank balance sheets, spreads, and sensitivities to rates.

3.10 Difference-in-differences validation of the reversal region

To validate the low-rate reversal logic in reduced form, the paper estimates

$$y_{i,t+4} = \beta(\mathrm{HHI}_i \times Low_t) + \gamma x_{i,t} + \eta_i + \eta_t + \varepsilon_{i,t}, \tag{34}$$

where $y_{i,t+4}$ is future log bank equity or log lending, HHI_i is local deposit-market concentration, and Low_t equals one when the federal funds rate is below 2%.

- **Interpretation.** If the reversal region is driven by deposit-market power, then high-HHI banks should be hit harder when rates are very low.
- **Role in the paper.** This is not part of structural estimation. It is an external validation exercise for the mechanism behind the reversal rate.

4 Identification and instruments (what makes the estimates credible)

4.1 Identification of deposit and loan demand

- **Core challenge.** Deposit and loan rates are endogenous because banks observe demand shocks (ξ_j^d, ξ_j^l) and set prices strategically.
- **Main instruments.** Salaries and noninterest expenses related to fixed assets serve as supply shifters, following Ho and Ishii (2011).
- **Exclusion restriction.** Conditional on observed product characteristics, bank fixed effects, and time fixed effects, these cost shifters must affect prices through supply rather than directly through borrower or depositor utility.
- **Market definition.** Each year is a market, and the estimation uses national markets so that rates and shares line up with available loan data.
- **Why time fixed effects matter.** Identification does *not* use monetary policy variation. Time fixed effects absorb aggregate shocks, which avoids the usual simultaneity problem between aggregate policy and aggregate credit supply.

4.2 What the first-stage parameters mean

- **Deposit rate sensitivity.** $\alpha^d = 0.968$: a 1 percentage point increase in the deposit rate raises a bank's market share by roughly 0.968%.
- **Borrower rate sensitivity.** $\alpha^l = -1.462$: a 1 percentage point increase in loan rates lowers a bank's market share by about 1.4%.
- **Depositor heterogeneity.** $\sigma_\alpha^d = 0.553$, which helps the model generate positive deposit spreads even in relatively unconcentrated markets.
- **Nonrate bank characteristics.** Branches and employees per branch significantly raise both deposit and loan demand, consistent with convenience and relationship-banking effects.

4.3 Identification in the second-stage SMD estimation

- **ϕ^N.** Identified mainly by the average nonreservable borrowing share. More costly wholesale funding implies more reliance on deposits.
- **ϕ^d and ϕ^l.** Identified by the average deposit and loan spreads.
- **ψ.** Identified using average net noninterest expenses and leverage. High fixed costs lower leverage and raise net noninterest expense.
- **γ.** Identified by the dividend yield, since impatient banks distribute more and retain less.
- **W/K and q_n^l.** Identified by the deposit-to-asset ratio and the sensitivity of aggregate credit to the federal funds rate.
- **Additional discipline.** The market-to-book ratio and the sensitivity of bank loans to the federal funds rate help ensure that the model matches the valuation of banks and the baseline strength of monetary transmission before the counterfactuals are run.

4.4 Low-dimensional equilibrium approximation

- **Computational problem.** A bank's true state depends on the cross-sectional distribution of all other banks, Γ_t, which is too high-dimensional to solve exactly.
- **Approximation.** Banks summarize the distribution of others' choices using average deposit and loan rates as functions of the current federal funds rate only.
- **Why believable?** The approximation performs well in simulation, with forecast R^2 values above 95%, and competition limits the extent to which one bank's dynamic distortions spill over to others.

4.5 External validation and causal design outside the structural estimation

- **Local projections with high-frequency instruments.** For external validation of impulse responses, the paper instruments the quarterly federal funds rate with high-frequency monetary-policy surprises around FOMC announcements.

- **FOMC-day stock-return design.** Monetary-policy news is measured by one-day changes in the two-year Treasury yield. The identifying assumption is that scheduled FOMC-window rate surprises primarily reflect policy news rather than contemporaneous macro fundamentals.
- **Low-rate DID.** The HHI $\times$ *Low* design compares banks with different local concentration when rates enter the low-rate region. This is meant to validate the mechanism that deposit market power matters especially when rates are near zero.

5 Tables and figures

5.1 Table I: Variable definitions

Read:

- The paper is careful about constructing *economy-wide* deposit and loan market sizes rather than using bank totals in isolation. That matters because demand shares are defined relative to outside options such as cash, Treasury bills, corporate debt, and equity.
- The spread variables are exactly matched to the model: the deposit spread is the federal funds rate minus the deposit rate, while the loan spread is the bank loan rate minus a five-year Treasury yield.
- The reserve ratio and nonreservable borrowing share are especially important because they map directly into the reserve channel and the balance-sheet-friction channel.

5.2 Table II: Summary statistics

Read:

- The banking sector is extremely skewed. Asset-weighted means for shares and branches sit far above equal-weighted means, so a few large banks dominate both deposits and lending.
- Average loan maturity around 3.4 years validates the paper's emphasis on maturity mismatch.
- Deposits fund a large share of the balance sheet, but nonreservable borrowing is still economically meaningful, which is exactly what allows balance-sheet frictions to matter.

Table I
Variable Definitions

Variable	Construction
Deposit market share	Deposits of a bank divided by the sum of deposits, cash, Treasury bills, and money market funds in the U.S. economy.
Loan market share	Loans of a bank divided by the sum of U.S. household debt, corporate debt, and corporate equity.
Deposit rates	Deposit interest expense divided by deposits.
Loan rates	Loan interest income divided by loans outstanding.
No. of branches	Number of local branches.
No. of employees per branch	Number of employees divided by number of branches.
Expenses related to fixed assets	Noninterest expenses related to the use of fixed assets divided by total assets.
Salary	Total salary expense divided by total assets.
Reserve ratio	10% times the weight of transaction deposits plus 1% times the weight of saving deposits.
Average loan maturity	Estimated maturity of each type loan weighted by the portfolio weight. Nonmortgage loan maturity is the repricing maturity and average prepayment adjusted mortgage duration comes from Landvoigt, Van Nieuwerburgh, and Elenev (2021).
Nonreservable borrowing share	Nonreservable borrowing divided by total deposits.
Deposit spread	Federal funds rate minus a deposit rate.
Loan spread	A loan rate minus the corresponding five-year Treasury yield.
Deposit-to-asset ratio	Deposits divided by total assets.
Net noninterest expense	Noninterest expense minus noninterest income, divided by total assets.
Leverage	Total assets divided by the book value of equity.
Market-to-book ratio	The market value of equity divided by the book value of equity.

Table II
Summary Statistics

In this table, we report summary statistics for our sample. The sample period is 1994 to 2017. The total size of the deposit market is defined as the sum of deposits, cash, and Treasury bills held by all U.S. households and nonfinancial corporations. The total size of the loan market is defined as the sum of U.S. corporate and household debt. Deposit and loan rates are calculated using interest expense and income. Expenses related to fixed assets and salaries are scaled by total assets. Deposit shares, loan shares, deposit rates, loan rates, expenses related to fixed assets, salaries, and net noninterest expenses are reported in percentages. Asset maturity is reported in years. "(vw)" indicates an asset-weighted mean, and "(ew)" indicates an equal-weighted mean. The data sources are the Call Reports and the FDIC Summary of Deposits.

	mean(vw)	mean(ew)	*SD*	p10	p25	p50	p75	p90
Deposit market shares	3.519	0.079	0.523	0.003	0.005	0.009	0.021	0.077
Loan market shares	1.368	0.033	0.207	0.001	0.002	0.004	0.009	0.034
Deposit rates	1.706	2.032	1.292	0.166	0.873	2.085	3.150	3.714
Loan rates	5.935	6.921	1.725	4.540	5.599	6.959	8.286	9.061
No. of branches	1778	69.753	315.678	7.000	11.000	17.000	34.000	94.000
No. of employees per branch	53.736	18.338	17.433	9.109	11.188	14.306	19.556	28.500
Expenses of fixed assets	0.454	0.480	0.165	0.270	0.347	0.448	0.584	0.798
Salaries	1.590	1.725	0.486	1.061	1.348	1.650	2.036	2.646
Net noninterest expenses	1.230	2.778	0.830	1.904	2.246	2.653	3.142	3.743
Loan-to-deposit ratio	0.816	0.815	0.170	0.598	0.710	0.821	0.925	1.022
Borrowing-to-deposit ratio	0.699	0.136	0.138	0.013	0.041	0.096	0.181	0.308
Deposit-to-asset ratio	0.707	0.805	0.082	0.691	0.763	0.822	0.866	0.895
Book leverage	11.464	11.114	2.577	7.947	9.408	10.990	12.656	14.390
Asset maturity	3.429	3.772	1.402	2.163	2.764	3.560	4.604	5.698

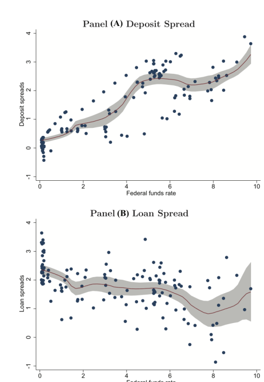

Figure 1. Deposit spread, loan spread, and the federal funds rate. In this figure, we plot kernel regressions of average deposit and loan spreads for U.S. banks on the federal funds rate. We use an Epanechnikov kernel with a bandwidth of 0.66 for deposits and 0.61 for loans. The sample period is 1985 to 2017. The data frequency is quarterly. The deposit and loan rates are constructed using the Call Reports, and the federal funds rate is from the FRED database. (Color figure can be viewed at wileyonlinelibrary.com)

5.3 Figure 1: Deposit spreads, loan spreads, and the federal funds rate

Read:

- Panel A shows deposit spreads rising with the federal funds rate, and the slope steepens near zero. This is the visual signature of deposit market power interacting with the cash outside option.
- Panel B shows loan spreads falling with the federal funds rate, consistent with banks cutting loan markups when funding conditions tighten.
- This figure is the raw reduced-form motivation for the whole model: policy rates do not pass through one-for-one to either side of the bank balance sheet.

5.4 Figure 2: Timeline within a period

Read:

- The timing is important: banks observe the federal funds rate and loan charge-offs, set deposit and loan rates, attract deposits and loans, choose reserves, securities, and wholesale borrowing, then collect profits and distribute dividends.
- This timing clarifies why the model is dynamic even though households and firms solve static demand problems. The dynamics come from the bank balance sheet, not from dynamic household or firm optimization.

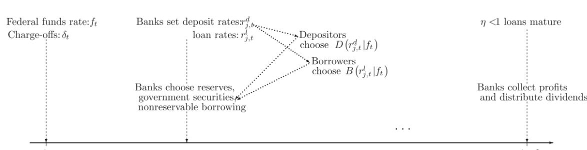

Figure 2. Timeline within a period.

Table III

Parameter Estimates

In this table, we report the model parameter estimates. Panel A presents calibrated parameters. Panel B presents values for parameters that can be calculated as simple averages or with simple regression methods. Panel C presents results for parameters estimated via BLP. Panel D presents results for parameters estimated via SMD. Standard errors for the estimated parameters are clustered at the bank level and reported in brackets.

	Panel A: Calibrated Parameters		
τ_c	Corporate tax rate	0.35	
θ	The reserve ratio	0.024	
κ	The capital ratio	0.06	
$\hat{J}$	Number of representative banks	6	
	Panel B: Parameters Estimated Separately		
μ	Average loan maturity	3.448	[1.445]
$\bar{f}$	Log Federal funds rate mean	−4.305	[0.320]
σ_f	Std of Federal funds rate innovation	0.551	[0.723]
ρ_f	Log Federal funds rate persistence	0.900	[0.040]
$\bar{\delta}$	Log loan chargeoffs mean	−5.905	[0.177]
σ_δ	Std log loan chargeoffs innovation	0.960	[0.339]
ρ_δ	Log loan chargeoffs persistence	0.600	[0.055]
$\rho_{\delta f}$	Corr of Federal funds rate innovation and log loan chargeoffs	−0.110	[0.069]
	Panel C: Parameters Estimated via BLP		
α^d	Depositors' sensitivity to deposit rates	0.968	[0.140]
σ_{α^d}	Dispersion of depositors' sensitivity to deposit rates	0.553	[0.116]
α^l	Borrowers' sensitivity to loan rates	−1.462	[0.292]
q^d_d	Convenience of holding deposits	3.440	[0.251]
q^d_c	Convenience of holding cash	1.985	[0.242]
q^l_l	Convenience of borrowing through loans	1.151	[1.065]
	Panel D: Parameters Estimated via SMD		
γ	Banks' discount rate	0.048	[0.007]
W/K	Relative size of the deposit market	0.217	[0.011]
q^l_n	Value of firms' outside option	−9.631	[0.262]
ϕ^N	Quadratic cost of nonreservable borrowing	0.010	[0.001]
ϕ^d	Bank's cost of taking deposits	0.010	[0.001]
ϕ^l	Bank's cost of servicing loans	0.007	[0.001]
ψ	Net fixed operating cost	0.027	[0.006]

5.5 Table III: Parameter estimates

Read:

- **Demand parameters.** Depositors are positively rate sensitive ($\alpha^d = 0.968$), borrowers are negatively rate sensitive ($\alpha^l = -1.462$), and depositor heterogeneity is sizable ($\sigma^d_\alpha = 0.553$).
- **Regulatory and calibrated parameters.** The reserve ratio is 0.024, the capital ratio is 0.06, and the model uses $\hat{J} = 6$ representative banks.
- **Balance-sheet frictions.** $\phi^N = 0.010$ implies that at the average wholesale-funding share of roughly 30%, an additional dollar of nonreservable borrowing costs about 30 basis points above the federal funds rate.
- **Operating costs.** $\phi^d = 0.010$, $\phi^l = 0.007$, and fixed net operating costs satisfy $\psi = 0.027$.
- **Discounting and outside options.** Banks' estimated discount rate is 4.8%, and the value of the firm's outside option not to borrow is very low ($q^l_n = -9.631$).

5.6 Table IV: Moment conditions

Read:

- The model matches the targeted moments closely. Simulated deposit and loan spreads are essentially identical to the empirical moments, and the loan-FFR sensitivity is -1.641 in the model versus -1.592 in the data.
- The fit to quantities is also good: the model matches the deposit-to-asset ratio, leverage, and market-to-book ratio reasonably well.
- This table matters because the later counterfactuals are only persuasive if the baseline model already reproduces the main balance-sheet and pricing facts in the data.

Table IV

Moment Conditions

In this table, we report simulated versus actual moments in the SMD estimation, along with t-statistics for the pairwise differences. The dividend yield is defined as dividends over bank equity value; the nonreservable borrowing share is defined as the ratio of nonreservable borrowing to total assets; the sensitivities of total credit and bank loans to the federal funds rate (FFR) are estimated via a vector autoregression.

	Actual Moment	Simulated Moment	t-statistic
Dividend yield	3.38%	2.87%	−0.856
Nonreservable borrowing share	29.90%	25.90%	−1.913
Std of nonreservable borrowing	12.60%	14.73%	0.818
Deposit spread	1.29%	1.32%	0.328
Loan spread	2.03%	2.04%	0.066
Deposit-to-asset ratio	0.699	0.737	1.034
Net noninterest expenses	1.20%	1.03%	−1.654
Leverage	11.20	11.89	1.398
Market-to-book ratio	2.061	1.796	−1.094
Credit-FFR sensitivity	−0.995	−1.008	−0.100
Bank loan-FFR sensitivity	−1.592	−1.641	−0.129

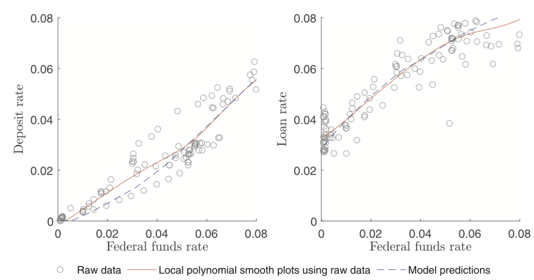

Figure 3. Model-predicted versus actual rates. This figure illustrates the relation between the federal funds rate and banks' deposit and loan rates. The circles represent a scatter of the raw data from 1994 to 2017, aggregated at the quarterly frequency. The dashed lines represent local polynomial smoothed plots based on the raw data. The solid lines represent the relations predicted using the model. (Color figure can be viewed at wileyonlinelibrary.com)

5.7 Figure 3: Model-predicted versus actual rates

Read:

- Both deposit and loan rates respond less than one-for-one to the federal funds rate, in both the data and the model.
- The model traces the nonlinear shape of the data rather well, especially the limited pass-through at low rates.
- This is strong external validation because these full response curves are *not* directly targeted in the moment-matching stage.

5.8 Figure 4: Bank concentration and monetary transmission

Read:

- Panel A shows that higher concentration lowers the sensitivity of deposit rates to policy but raises the sensitivity of deposit *quantities*. More market power means banks can move spreads more aggressively when policy changes.
- Over the interquartile range of HHI, the model predicts about a 3 basis point rise in deposit-spread sensitivity and about a 70–76 basis point rise in deposit-quantity sensitivity, close to Drechsler, Savov, and Schnabl (2017).
- Panel B shows that more concentration dampens pass-through to loan rates and quantities. This lines up with Scharfstein and Sunderam's mortgage-market evidence.
- A subtle but important point is that deposit-spread sensitivity stays positive even at low HHI, because depositor heterogeneity itself creates market power even without extreme concentration.

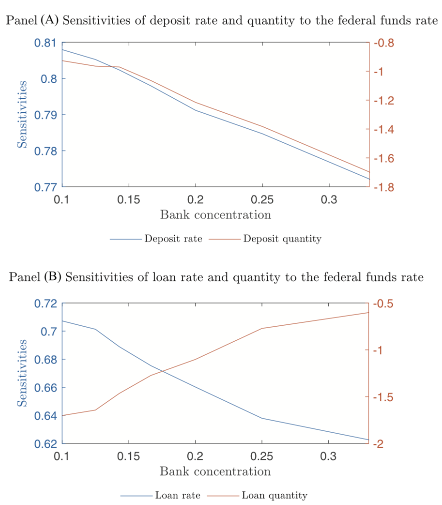

Figure 4. Bank concentration and monetary transmission. This figure plots model-generated sensitivities of deposit and loan rates and quantities to the federal funds rate. Panel A plots the deposit rate and deposit quantity sensitivities, each as a function of market concentration, which we measure as the market Herfindahl index (HHI). Panel B plots the loan rate and loan quantity sensitivities, each as a function of the HHI. We calculate rate sensitivities by regressing changes in the deposit or loan rate on changes in the federal funds rate. We calculate quantity sensitivities by regressing changes in the log quantity on changes in the federal funds rate. (Color figure can be viewed at wileyonlinelibrary.com)

Table V

Sensitivities of Additional Moments to Monetary Shocks

This table reports the sensitivities of bank deposit spreads, loan spreads, noninterest income, assets, loans, securities, deposits, and borrowing to the federal funds rate. In the second column, we report the sensitivities estimated in the data. The data are from the Call Reports and include all U.S. commercial banks from 1994 to 2008 at a quarterly frequency. We estimate impulse responses to a 1% change in the federal funds rate using the local projections method of Jordà (2005) at an eight-quarter horizon. We instrument the federal funds rate with the unexpected monetary policy shocks constructed from changes in the federal funds futures price on FOMC announcement dates, as in Bernanke and Kuttner (2005). We follow Gertler and Karadi (2015) and English, Van den Heuvel, and Zakrajšek (2018) to sum up the surprises within a quarter to form the instrument. In the third column, we report the sensitivities predicted by the model, which correspond to changes in the variables of interest over a two-year horizon scaled by their respective steady-state levels following a 1% change in the federal funds rate.

Panel A: Sensitivity of Prices		
	Empirical	Model
Deposit spread	0.386	0.310
Loan spread	−0.024	−0.011
Net noninterest expense	−0.001	0.000
Panel B: Sensitivity of Quantities		
	Empirical	Model
Deposit	−2.491	−2.482
Loan	−1.949	−1.641
Borrowing	−3.195	−5.335
Securities	−1.232	−3.052
Assets	−2.587	−2.773

5.9 Table V: Sensitivities of additional moments to monetary shocks

Read:

- The model does a good job matching the two-year responses of deposit spreads (0.310 model versus 0.386 empirical) and loan spreads (−0.011 versus −0.024).
- It also fits the responses of balance-sheet quantities reasonably well: deposits, loans, assets, and securities all move in the right directions and with similar magnitudes.
- An extra validation result in the text is that a 1 percentage point rate shock lowers bank equity values by 1.44% in the model versus 1.93% in the data. Without deposit market power, the model would overpredict this sensitivity dramatically.

5.10 Table VI: Determinants of monetary policy transmission

Read:

- In the baseline model with all frictions, a 1 percentage point increase in the federal funds rate lowers lending by 1.641% over two years.
- Removing reserve regulation changes that sensitivity only to −1.499, an 8.65% reduction in transmission. This is the paper's clearest evidence that the reserve channel is weak in the sample period.
- Removing capital regulation reduces the lending sensitivity to −1.248, a 23.91% decline in transmission.
- Removing deposit market power reduces the sensitivity further to −1.132, a 31.00% decline, making deposit market power the single strongest amplifying friction in the baseline decomposition.
- Removing loan market power pushes the sensitivity to −1.951, so loan market power actually softens the response of lending to monetary policy.

Table VI

Determinants of Monetary Policy Transmission

This table presents the results of a series of counterfactual experiments in which we examine the effects of removing frictions from our model. The first column lists the frictions that are removed from the model. The second column presents the sensitivity of loans to the federal funds rate (FFR) when the corresponding frictions are removed. The sensitivity captures changes in loans over a two-year horizon scaled by the steady-state loan-level following a 1% change in the federal funds rate. The third column presents the percent change in the sensitivity relative to the baseline case in row (1). All model solutions are under the same set of parameters reported in Table III.

		Sensitivity of Loans to FFR	Change Relative to Baseline (%)
(1)	All frictions are present	−1.641	/
(2)	− Reserve regulation	−1.499	8.65%
(3)	− Capital regulation	−1.248	23.91%
(4)	− Deposit market power	−1.132	31.00%
(5)	− Loan market power	−1.951	−18.91%

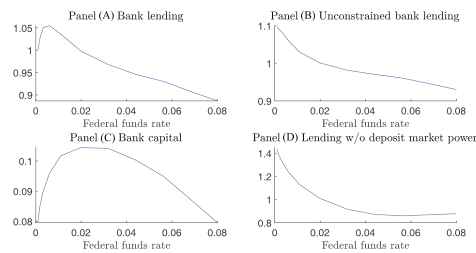

Figure 5. Bank capital, bank lending, and the federal funds rate. This figure illustrates how bank capital and optimal lending vary with the federal funds rate. In all panels, the federal funds rate is on the *x*-axis. Bank characteristics, scaled by the level of steady-state bank lending, is on the *y*-axis. (Color figure can be viewed at wileyonlinelibrary.com)

5.11 Figure 5: Bank capital, bank lending, and the federal funds rate

Read:

- Panel A shows the headline reversal-rate result: lending is hump-shaped in the federal funds rate, with the turning point around 0.9%.
- Panel B shows desired lending always falls with the policy rate, so the reversal is *not* coming from firms suddenly wanting fewer loans at lower rates.
- Panel C shows bank capital is itself hump-shaped in the rate, with a turning point around 2%. At very low rates, compressed deposit spreads reduce profits and thus capital accumulation.
- Panel D shows that if deposit market power is shut down, the reversal disappears. This figure therefore isolates the interaction between deposit market power and capital regulation as the mechanism.

5.12 Figure 6: Impulse responses to federal funds rate shocks

Read:

- Starting from the inflection point (0.9%), either a hike to 2% or a cut to 0.1% eventually lowers lending.
- After a hike, deposits fall sharply because banks widen deposit spreads; wholesale borrowing rises to fill the gap; lending declines gradually because loans are long-term.
- After a cut, deposits initially rise, and lending rises briefly, but the bank's capital position deteriorates over time because deposit-market profits are squeezed near the zero lower bound. Lending then falls back below its initial level.
- The figure is important because it shows that "lending falls after either sign of shock" does not mean the mechanisms are the same. Tightening works through funding costs and deposit outflows; easing at very low rates works through profitability and capital.

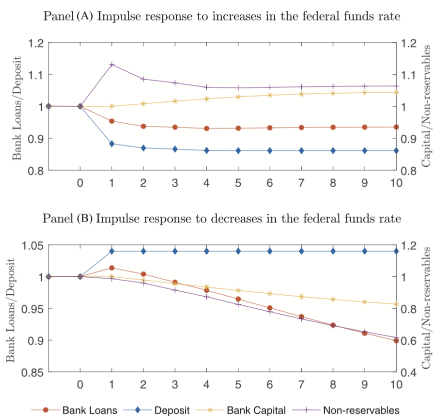

Figure 6. Impulse response to federal funds rate shocks. This figure illustrates banks' impulse responses to federal funds rate shocks. The economy starts at Year 0 when it is in the old steady state with the federal funds rate equal to 0.9%. In Year 1, the federal funds rate either increases to 2% or decreases to 0.1%, and it stays at that level afterward until the economy reaches the new steady state. Each variable in the graph is scaled by the level in the old steady state in which the federal funds rate is 0.9%. (Color figure can be viewed at wileyonlinelibrary.com)

Table VII

Monetary Policy Shocks and Bank Equity Returns on FOMC Days

In this table, we report the estimates of the relation between bank equity returns and monetary policy shocks on FOMC days. Monetary policy shocks are measured as one-day changes in the two-year Treasury yield on FOMC days. HHI is the Herfindahl-Hirschman Index for the local deposit market in which a bank operates. Low is a dummy variable that equals 1 when the starting level of the federal funds rate (FFR) is below 2%. The control variables include market returns and term spreads. The sample includes all publicly traded U.S. banks from 1994 to 2017. The sample for columns (1) and (4) comprises observations in which the starting level of the federal funds rate (FFR) is above 2%. The sample for columns (2) and (5) comprises observations in which the starting level of the federal funds rate is below 2%. The sample for columns (3) and (6) comprises all observations. We exclude observations during the collapse of the dot-com bubble (2000 to 2001) and the subprime financial crisis (2007 to 2009). Standard errors are clustered by time.

	High (1)	Low (2)	All (3)	High (4)	Low (5)	All (6)
Policy shock	-1.292^{**}	2.202^{**}	-1.292^{**}	-0.639	-1.393	-0.639
	[0.615]	[0.879]	[0.612]	[0.653]	[0.852]	[0.649]
Low*Policy shock			3.494^{***}			-0.754
			[1.069]			[1.069]
HHI*Policy shock				-0.134	0.562^{***}	-0.134
				[0.145]	[0.153]	[0.144]
Low*HHI*Policy shock						0.696^{***}
						[0.210]
Control	Yes	Yes	Yes	Yes	Yes	Yes
Observations	27,257	33,805	61,062	27,257	33,805	61,062
Adj. R^2	0.015	0.123	0.074	0.016	0.125	0.075

5.13 Table VII: Monetary policy shocks and bank equity returns on FOMC days

Read:

- In high-rate environments, the coefficient on the policy shock is negative (-1.292), so positive rate surprises lower bank equity returns.
- In low-rate environments, the coefficient flips sign to 2.202, and the pooled interaction term *Low* $\times$ *Policy shock* is strongly positive (3.494). This is reduced-form validation of the model's reversal logic for bank capital.
- The interaction with HHI is also telling: in low-rate environments, banks with more local deposit market power have more positive stock-return responses to rate increases.

5.14 Figure 7: Monetary policy shocks and bank equity returns

Read:

- The scatter plots visually summarize Table VII. The slope is negative when the federal funds rate starts above 2% and positive when it starts below 2%.

- This figure makes the nonmonotonicity immediately visible and helps the reader see that the sign flip is not driven by a handful of outliers.

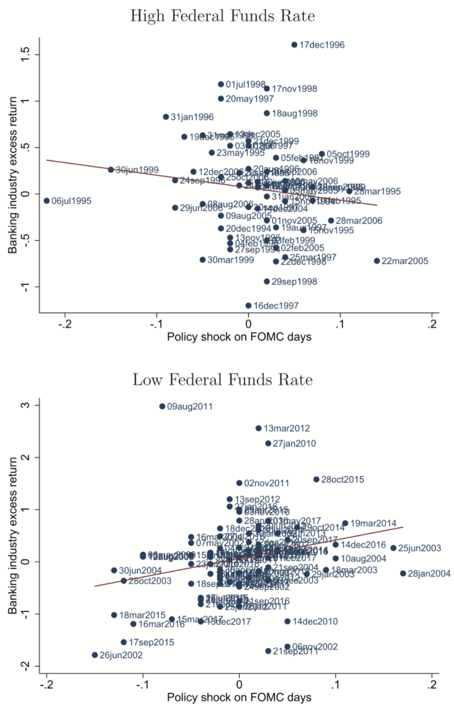

Figure 7. Monetary policy shocks and bank equity returns. This figure provides scatter plots of bank industry excess returns against monetary policy shocks on FOMC days from 1994 to 2017. The excess returns are defined as the difference between bank industry index returns and market returns. Monetary policy shocks are measured as one-day changes in two-year Treasury yields on FOMC days. The sample for the upper panel comprises observations in which the starting level of the federal funds rate is above 2%. The sample for the lower panel comprises observations in which the starting level of the federal funds rate is below 2%. We exclude observations during the collapse of the dot-com bubble (2000 to 2001) and the financial crisis (2007 to 2009). Bank industry stock returns are from Kenneth French's website, and the two-year Treasury yield is from the FRED database. (Color figure can be viewed at wileyonlinelibrary.com)

Table VIII
Effects of Low Rates on Banks with Different Deposit Market Power

We report the estimates of the effect of low interest rates on banks with different deposit market power. HHI is the average Herfindahl-Hirschman Index of the local deposit markets in which a bank operates. Low is a dummy variable that equals 1 when the federal funds rate is below 2%. Each column is identified by the dependent variable. The control variables include lagged assets, lending, deposits, equity, deposit rate, loan rate, bank fixed effects, and time fixed effects. The sample includes all U.S. banks from 2000Q1 to 2004Q1. Standard errors are clustered by time.

	Equity (1)	Equity (2)	Loan (3)	Loan (4)
HHI*Low	−0.232***	−0.083***	−0.252***	−0.055***
	(0.007)	(0.006)	(0.008)	(0.006)
Control	No	Yes	No	Yes
Bank F.E.	Yes	Yes	Yes	Yes
Time F.E.	Yes	Yes	Yes	Yes
Observations	129,950	127,912	129,339	127,885
Adj. R^2	0.984	0.990	0.981	0.990

5.15 Table VIII: Effects of low rates on banks with different deposit market power

Read:

- The interaction HHI × *Low* is negative for both equity and lending, with and without controls.
- In the controlled specification, low rates reduce equity more for high-HHI banks by 0.083 log points and reduce lending more by 0.055 log points.
- This is exactly the pattern the model predicts if very low rates are especially damaging for banks that rely more on deposit market power.

5.16 Table IX: Large banks versus small banks

Read:

- Small banks have much higher wholesale funding costs ($\phi^N = 0.015$ versus 0.006) and much larger fixed operating costs ($\psi = 0.075$ versus 0.013).

- Their lending is therefore much more sensitive to the federal funds rate: -1.742 for small banks versus -1.235 for large banks.
- The paper's counterfactuals suggest that roughly 47% of the difference in sensitivity can be explained by the external-finance-cost gap alone.
- This provides a structural microfoundation for the classic Kashyap–Stein idea that small banks are hit harder because they cannot cheaply replace deposits.

Table IX

Large Banks versus Small Banks

In this table, we report SMD estimation results for subsamples of large and small banks. Panel A contains the simulated versus actual moments, along with t-statistics for the pairwise differences. In Panel B, we report the parameter estimates. ϕ^d and ϕ^l are banks' marginal costs of intaking deposits and servicing loans, respectively, ψ is the net fixed operating cost, and ϕ^N is the quadratic cost of borrowing nonreservables. The last column presents the sensitivity of loans to the federal funds rate (FFR). Standard errors clustered at the bank level are in brackets under the parameter estimates.

Panel A: Moment Conditions

	Large Banks			Small Banks		
	Actual	Simulated	t-statistic	Actual	Simulated	t-statistic
Dividends	3.60%	2.89%	−3.568	/	/	/
Nonreservable borrowing share	35.50%	36.66%	0.612	15.30%	16.09%	1.574
Std of nonreservable borrowing	13.30%	19.86%	2.626	8.70%	10.83%	2.363
Deposit spread	1.32%	1.37%	0.520	1.25%	1.26%	0.130
Loan spread	1.77%	1.98%	2.068	2.71%	2.42%	−2.979
Deposit-to-asset ratio	0.666	0.688	0.692	0.784	0.779	−0.382
Net noninterest expense	0.96%	0.85%	−1.117	1.90%	1.76%	−1.436
Leverage	11.36	12.54	2.026	10.78	10.03	−3.796
Market-to-book ratio	2.06	1.97	−1.217	/	/	/
Total credit-FFR sensitivity	−0.995	−0.998	−0.010	−0.995	−1.027	−0.108

Panel B: Parameter Estimates

	ϕ^N	ϕ^d	ϕ^l	ψ	W/K	Sensitivity of Loans to FFR
Large banks	0.006	0.010	0.006	0.013	0.228	−1.235
	[0.001]	[0.001]	[0.001]	[0.003]	[0.007]	
Small banks	0.015	0.008	0.009	0.075	0.140	−1.742
	[0.001]	[0.001]	[0.001]	[0.002]	[0.001]	

Table X

Subsample Estimates: Early versus Late

In this table, we report the model parameter estimates for the early (1994 to 2005) and late (2006 to 2017) subsamples. Panel A presents calibrated parameters. Panel B presents values for parameters that can be calculated as simple averages or with simple regression methods. Panel C presents results for parameters estimated via BLP. Panel D presents results for parameters estimated via SMD. Standard errors for the estimated parameters are clustered at the bank level and reported in brackets.

		Early Subsample		Late Subsample	
Panel A: Calibrated Parameters					
τ_c	Corporate tax rate	0.350		0.350	
θ	The reserve ratio	0.028		0.022	
κ	The capital ratio	0.060		0.060	
$\hat{J}$	Number of representative banks	7		5	
Panel B: Parameters Estimated Separately					
μ	Average loan maturity	3.170	[1.402]	3.590	[1.448]
$\bar{f}$	Log Federal funds rate mean	−3.305	[0.170]	−5.605	[0.416]
σ_f	Std of log federal funds rate innovation	0.553	[0.148]	0.516	[0.885]
ρ_f	Log Federal funds rate persistence	0.700	[0.141]	0.900	[0.111]
$\bar{\delta}$	Log Loan chargeoffs mean	−6.005	[0.163]	−5.806	[0.258]
σ_δ	Std of log loan chargeoffs innovation	0.800	[0.255]	1.040	[0.383]
ρ_δ	Log Loan chargeoffs persistence	0.600	[0.060]	0.600	[0.068]
$\rho_{\delta f}$	Corr of federal funds rate innovation and chargeoffs	0.040	[0.435]	−0.160	[0.159]
Panel C: Parameters Estimated via BLP					
α^d	Depositors' sensitivity to deposit rates	0.743	[0.165]	0.925	[0.399]
σ_{α^d}	Dispersion of depositors' sensitivity to deposit rates	0.424	[0.144]	0.528	[0.279]
α^l	Borrowers' sensitivity to loan rates	−1.017	[0.054]	−1.454	[0.082]
q_d^d	Convenience of holding deposits	3.465	[0.358]	2.340	[0.470]
q_c^d	Convenience of holding cash	2.763	[0.387]	−0.444	[0.430]
q_l^l	Convenience of borrowing through loans	−0.016	[0.088]	1.804	[0.212]
Panel D: Parameters Estimated via SMM					
γ	Banks' discount rate	0.047	[0.002]	0.044	[0.005]
W/K	Relative size of the deposit market	0.184	[0.005]	0.254	[0.007]
ϕ^N	Quadratic cost of nonreservable borrowing	0.010	[0.001]	0.010	[0.004]
ϕ^d	Bank's cost of taking deposits	0.009	[0.001]	0.009	[0.004]
ϕ^l	Bank's cost of servicing loans	0.005	[0.001]	0.008	[0.002]
ψ	Net fixed operating cost	0.048	[0.002]	0.010	[0.005]

5.17 Table X: Early versus late subsample estimates

Read:

- Monetary transmission weakens over time: the lending sensitivity declines from about -1.53 in 1994–2005 to about -1.22 in 2006–2017.
- Concentration increases over time ($\hat{J}$ falls from 7 to 5), but at the same time depositors and borrowers become more rate-sensitive, which works against rising market power.
- Fixed operating costs fall sharply (ψ from 0.048 to 0.010), and the average federal funds rate falls substantially, making the economy spend more time near the reversal region.
- The decomposition in the text attributes about 34% of the weakening in transmission to changes in market-power parameters, 22% to operating and funding frictions, and the remaining 44% to macro conditions, especially the lower average policy rate.

6 Short summary

- **Conclusion.** The paper shows that bank market power—especially in the deposit market—is a first-order determinant of monetary policy transmission, with a quantitative importance comparable to bank capital regulation and far larger than reserve requirements in the sample period.
- **Intuition.** Monetary policy does not pass through mechanically. Banks choose how much of a funding-cost shock to transmit to depositors and borrowers, and that choice depends on market power, funding frictions, and capital constraints.
- **Empirical lesson.** To understand bank lending responses to policy, one needs a framework that combines industrial-organization demand estimation with dynamic balance-sheet constraints. Reduced-form evidence alone cannot cleanly rank the competing channels.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Deposit market power amplifies monetary policy transmission through banks. Capital regulation also matters a lot, but reserve requirements matter relatively little over 1994–2017.
- **Mechanism.** When rates rise, banks widen deposit spreads, deposits shrink, and costly non-reservable borrowing limits how much they can maintain lending. When rates fall very low, deposit spreads compress, profits weaken, retained earnings slow, and the capital constraint tightens.
- **Quantitative ranking.** Removing deposit market power reduces the lending response by 31%, removing capital regulation reduces it by 24%, and removing reserve regulation reduces it by only 9%.
- **Cross-sectional implication.** Small banks are more exposed because wholesale finance and fixed operating costs are more burdensome for them.
- **Time-series implication.** The post-2006 weakening in transmission partly reflects lower average rates that place the economy closer to the reversal region.
- **External validation.** The model matches rate pass-through, balance-sheet responses, equity-value responses, concentration patterns, and the sign flip in bank stock returns at low rates.

7.2 Economic intuition

- Banks are not price takers. Their ability to hold down deposit rates and smooth loan rates is itself a transmission channel.
- Deposit-market power matters because it changes the *quantity* of deposits available to support lending. Once wholesale funding is costly, that quantity effect becomes a lending effect.

- Capital regulation and market power interact. Near zero rates, the usual expansionary effect of lower funding costs can be overwhelmed by the effect of compressed deposit spreads on profits and capital accumulation.

7.3 Takeaway

This paper's main contribution is to put market structure at the center of monetary transmission through banks. In the authors' estimated model, bank market power is not a side detail; it is one of the main reasons policy-rate changes show up in lending. The most striking implication is the reversal-rate result: once rates are very low, further cuts can backfire because banks lose the deposit-market rents that help them sustain capital and lending.